

Instruments & Systems
Valve Automation & Control Technology

INSTRUMENTS & SYSTEMS

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TEMPERATURE TRANSMITTER

DUAL INPUT

Operating & Maintenance Manual



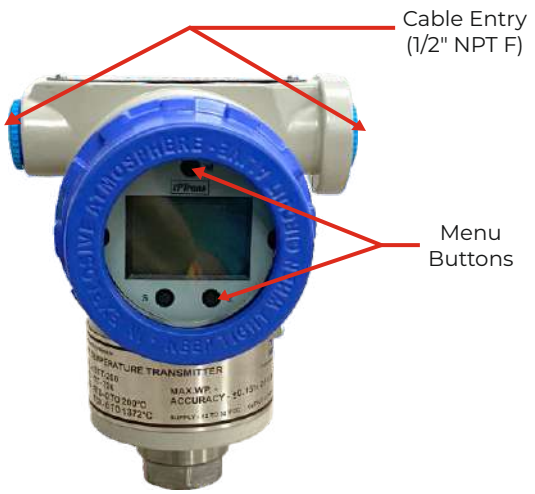


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30 Mohabewala Industrial Area,
Mohabewala, Dehradun,
(Uttarakhand) - 248002

OUR COMPANY PROFILE

1. General Information

Front View



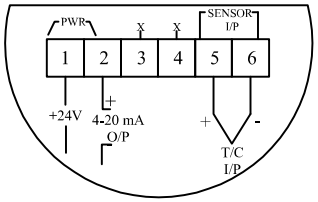
Back View (Single Input)



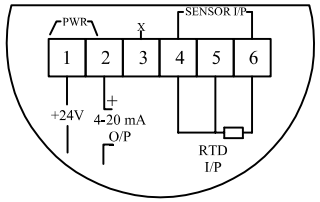
2. Electrical Connection

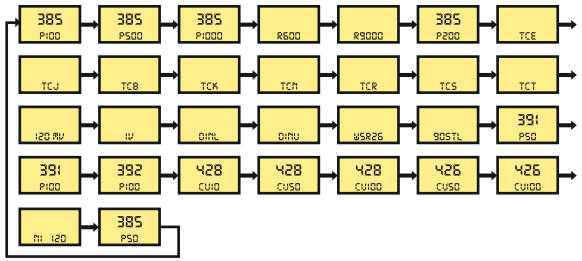
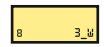
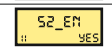
(i) For Single Input

Single Input Thermocouple



Single Input RTD



Manual Numbers		Description	Default Value in Display
0	0	No Change	
1	1	No Change	
2	2	Units : °C, °F, °R, K, Mv and OHM	
3	3	Low PV eg : 0.000	
4	4	High PV eg : 200.00	
5	5	Damping eg : 0.0000	
6	6	Zero Setting eg : 0.0000	
7	7	Input Type Connection 	
8	8	Number of wire connection : 2_W, 3_W & 4_W eg : 3 wire	
9	9	OHM	
11	11	Second Sensor enabling/disabling	
12	12	Hot backup enabling/disabling	
13	13	Dual Sensor Function Temperature Difference DIFFT Average Temperature AVERT Main Sensor Priority F_S_T	

Note :-

When Sensor 1 fails, E1 is indicated on the bottom left side of the display and E3 is indicated when sensor 2 fails.

Menu 8: Input for Number of Wire Connection

1. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
2. Key Z- Again press this key for 4 times- '00000'
3. Key S- Press 8 times- '00008'
4. Key M- Press 1 time to go to Menu-8.
5. Key S- Press this key to select wire connection as 2 Wire, 3 Wire & 4 Wire.
6. Key M- Press this key until we get the 0 Menu.
7. Key S- Lastly press 1 time to save the setting.

Number of wire connection : 2_W, 3_W & 4_W eg : 3 wire	

6. GA Drawing & Dimensions

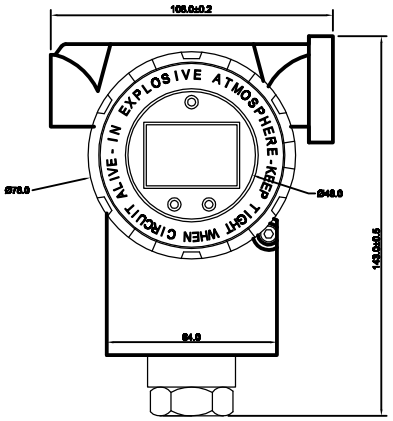


FIG.- FRONT VIEW

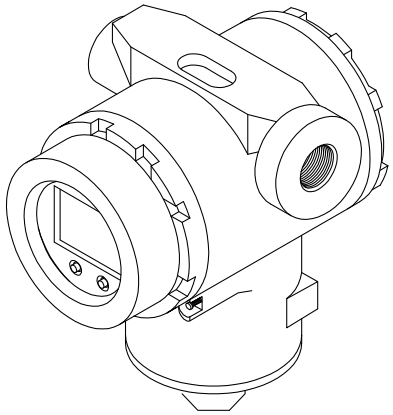
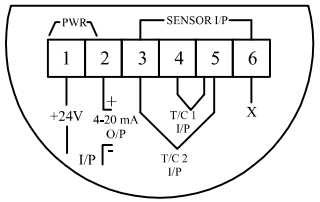


FIG.- ISOMETRIC VIEW

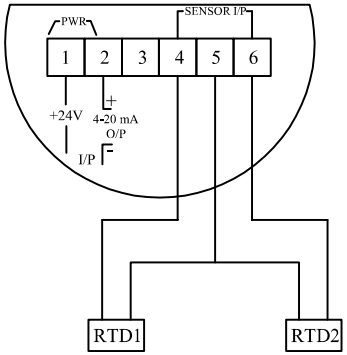
2. Electrical Connection

(ii) For Dual Input

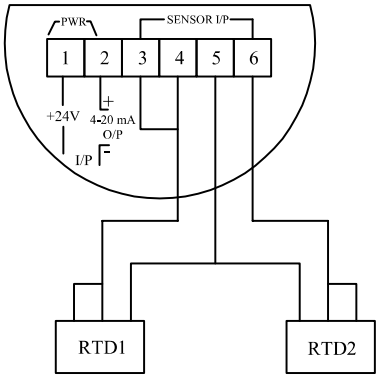
Dual Input Thermocouple



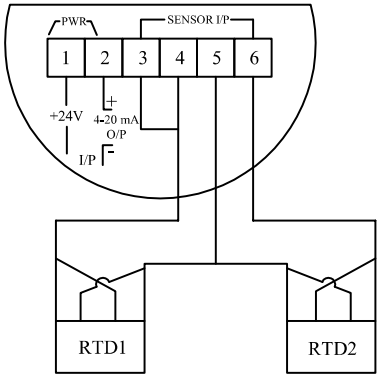
Dual Input RTD 2-Wire



Dual Input RTD 3-Wire

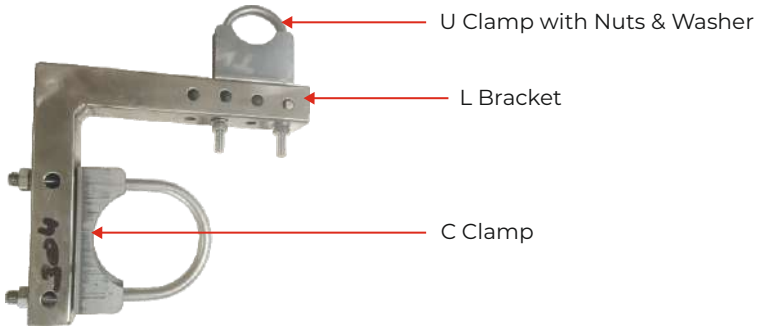


Dual Input RTD 4-Wire



3. Mechanical Installation

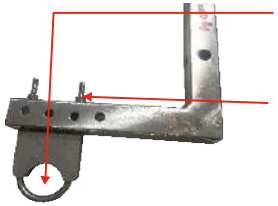
(Vertical / Horizontal)



STEP 1



Place the C Clamp with the L Bracket then ahead of C Clamp put transmitter and fasten it with U Clamp using nuts & washers.



Note :

1. The smaller size U Clamp & C Clamp pair will be used for PT mounting.

2. The bigger size U Clamp & C Clamp pair will be used for Pipe/Panel.

3. The clamping should be done with L Bracket in vertical or horizontal holes as per the mounting pipe requirement.

STEP 2



1. Now clamp the other side of the bracket as the same.

2. Place the C Clamp with the L Bracket then ahead of C Clamp and tighten it with the pipe/panel with U Clamp using nuts & washers tightly.



- The mounting arrangement of the PT in panel or pipe would be like this.

- This is the vertical pipe mounting it can be done with horizontal pipe mounting also, we just have to clamp the U Clamp in vertical holes of the L Bracket then the mounting will be as per horizontal pipe mounting.

Please note :-

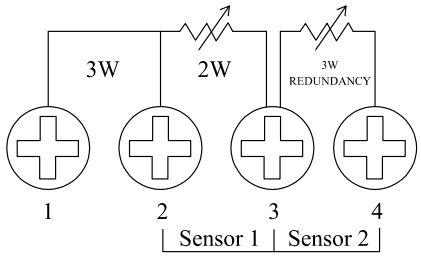
1. for vertical pipe mounting the both U Clamp & C Clamp should be tighten in the horizontal holes.

2. for horizontal pipe mounting the both U Clamp & C Clamp should be tighten in the vertical holes.

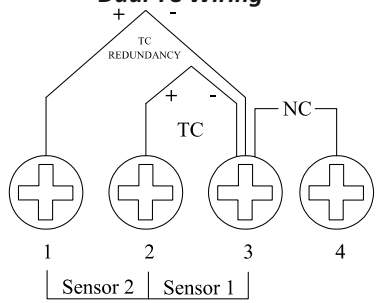
5. Sensor Connection/Wiring Diagram

Through variable “Enable/Disable the second sensor” confirm whether to connect dual sensor. If disable of the second sensor, the wiring method is same , if enable of the second sensor, the wiring method as below :-

Dual RTD Wiring



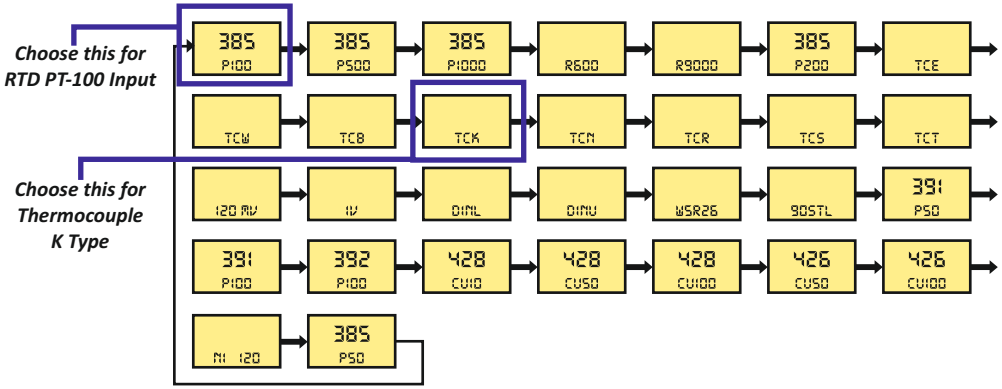
Dual TC Wiring



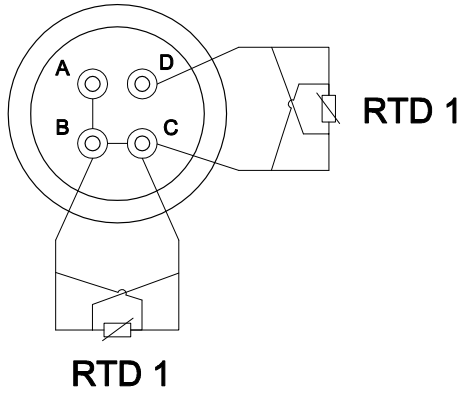
6. Adjustments with Keypad

Menu 7: For Input Type Connection Setting (RTD/TC)

1. Key Z- Long press for 2-3 seconds. LCD will display a code- '00000'
2. Key Z- Again press this key for 4 times- '00000'
3. Key S- Press 7 times- '00007'
4. Key M- Press 1 time to go to Menu-7.
5. Key S- Press this key until we get the required input type connection.
6. Key M- Press this key until we get the 0 Menu.
7. Key S- Lastly press 1 time to save the setting.



Dual Input RTD 4-Wire



FEMALE PART

LOOP DIAGRAM:-

A - RTD 1

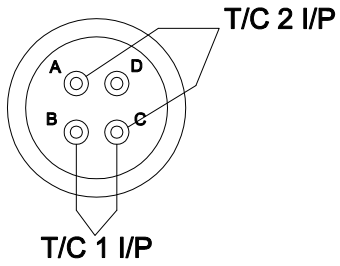
B - RTD 1

C - COM

D- RTD 2

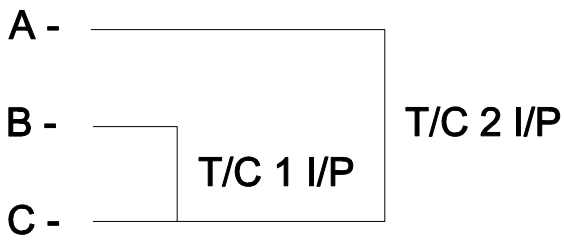
4. Wiring Diagram

Dual Input Thermocouple



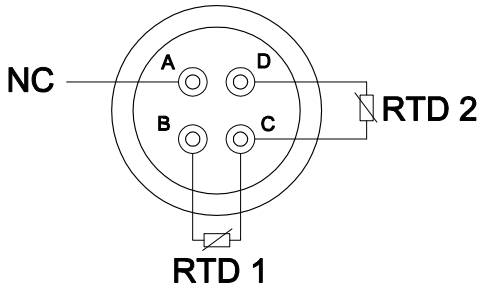
FEMALE PART

LOOP DIAGRAM:-



D - NC (NO CONNECTION)

Dual Input RTD 2-Wire



FEMALE PART

LOOP DIAGRAM:-

A - NC (NO CONNECTION)

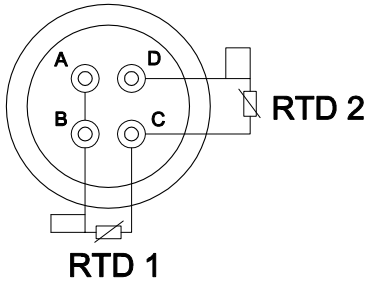
B - RTD 1

C - COM

D- RTD 2

TEMPERATURE TRANSMITTER
OPERATING & MAINTENANCE MANUAL

Dual Input RTD 3-Wire



FEMALE PART

LOOP DIAGRAM:-

A -
B - RTD 1

C - COM

D- RTD 2

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